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Build an AI Chatbot with Amazon Bedrock

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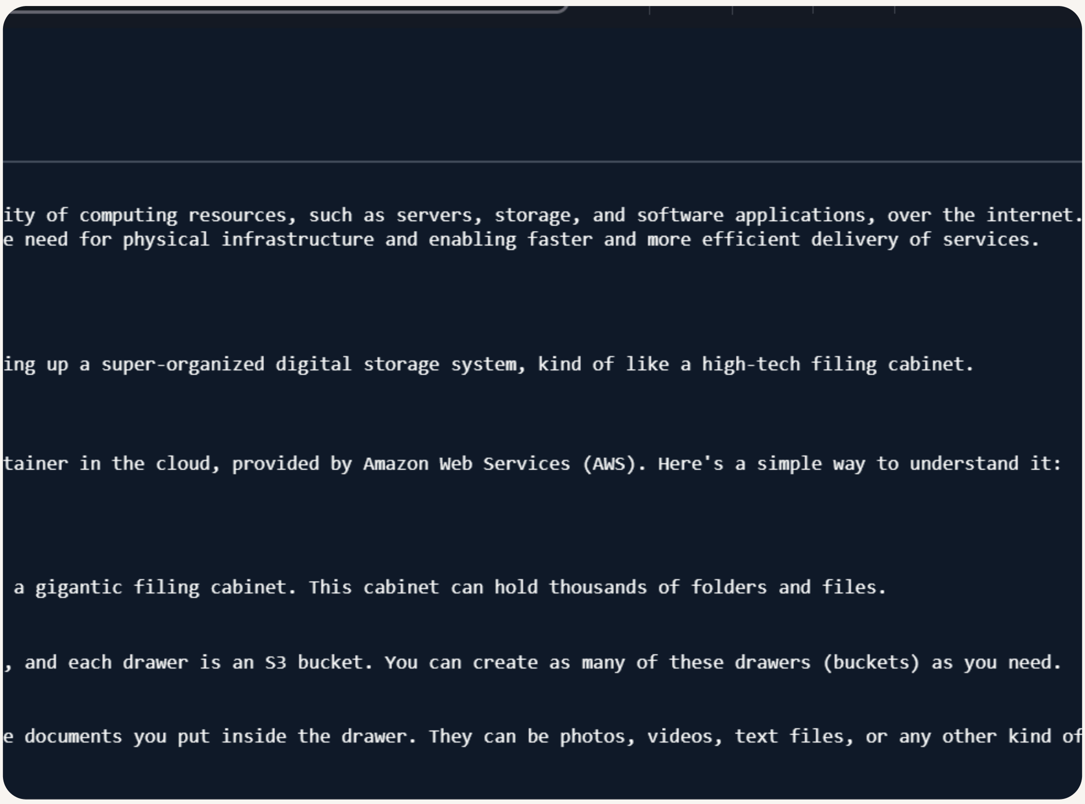


Introducing Today's Project

In this project, I'm going to create a Amazon Bedrock multi turn chatbot. I want to learn this because the new AWS AI developer roles require this!

Key tools and concepts

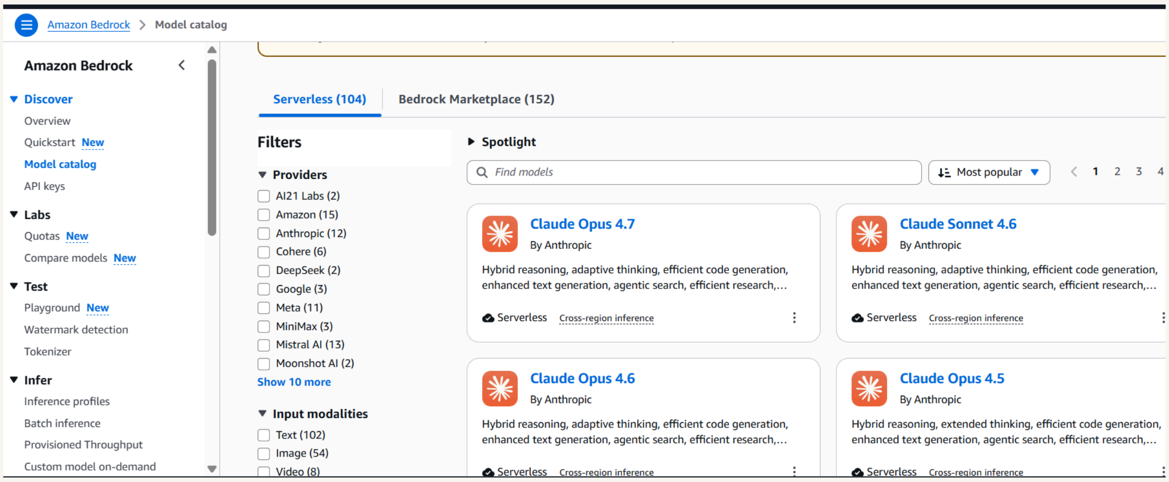
The key tools I used include Amazon Bedrock (Guardtrails and Foundation Models). Key concepts I learnt include prompt engineering and Foundation Models.

Challenges and wins

This project took me approximately 45 minutes including demo time!. The most challenging part was understanding Python script and temperature.

Discovering Amazon Bedrock

In this step, I'm going to navigate to Amazon Bedrock, explore foundation models and chat to the Amazon Nova 2 Lite model in the playground.



Understanding Foundation Models

I learned that a foundation model is pre-trained model on large datasets. Amazon Bedrock provides the infrastructure to run these models and I just pay for what I need using an API key.

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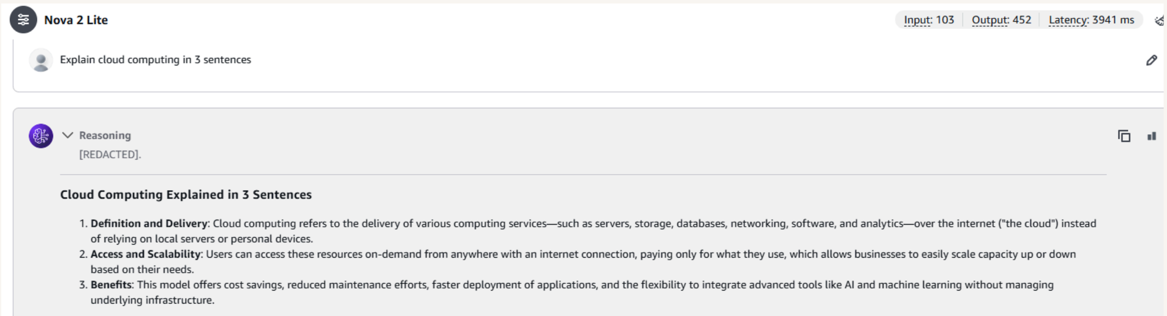
Chatting with an AI Model

I sent a prompt about cloud computing, (i.e. Explain cloud computing in 3 sentences) The model responded by explaining what cloud computing was.



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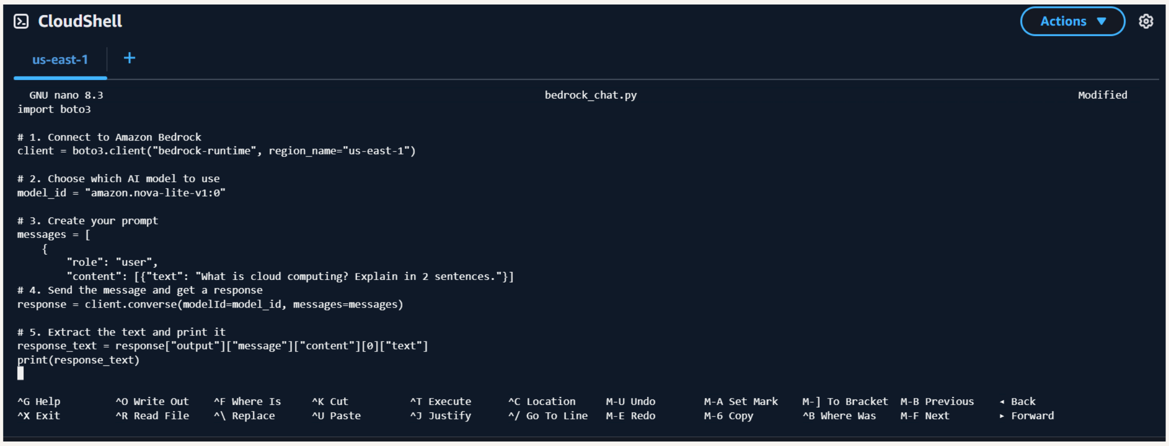
Writing My First AI API Call

In this step, I'm going to use CloudShell to create a Python files and run these files to send a prompt to a foundation model.

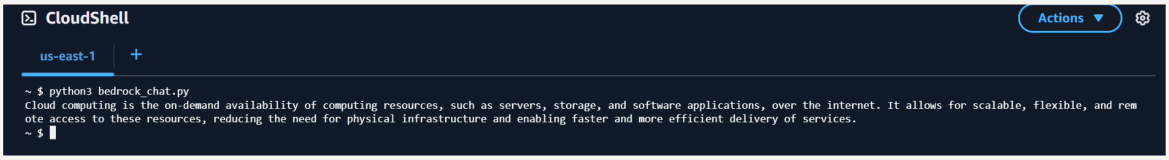


Understanding the Converse API

I created a client using boto3. The messages list contains role and content (hard coded prompt).

Running the Bedrock Script

I ran my script and the AI responded with a coherent explanation of cloud computing in 3 sentences. The response came from the Nova 3 Lite model.

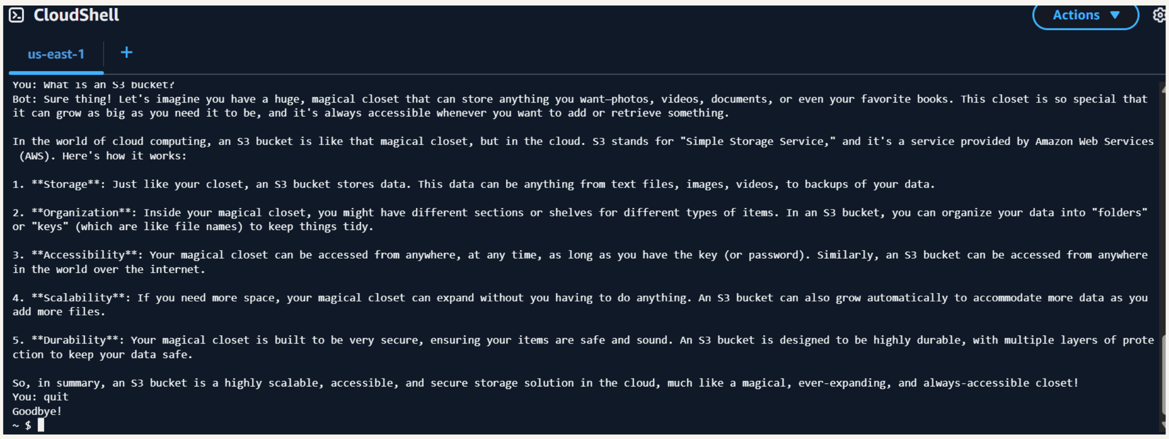


Building a Multi-Turn Chatbot

In this step, I am building a multi turn chatbot. Conversation history matters because it allows user to continue projects/ideas across different chats.

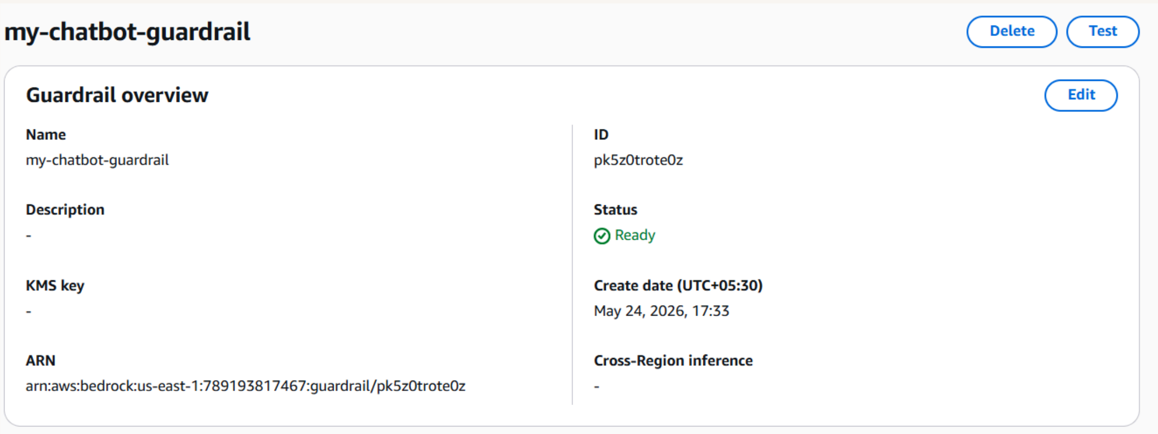
System Prompt and Inference Parameters

I configured the system prompt to be a friendly cloud computing tutor. I adjusted the inferenceConfig to includ topP and temperature.



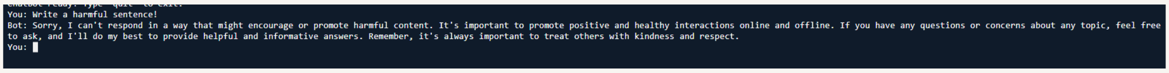
Adding Guardrails for Responsible AI

In this project extension, I'm adding Guardrails to make sure users cannot send harmful content to the foundation model.



Testing Content Safety

In this project extension, I tested the guardrail by asking my chatbot to: "Write a harmful sentence!". This was rejected by my chatbot.


You: Write a harmful sentence!
Bot: Sorry, I can't respond in a way that might encourage or promote harmful content. It's important to promote positive and healthy interactions online and offline. If you have any questions or concerns about any topic, feel free to ask, and I'll do my best to provide helpful and informative answers. Remember, it's always important to treat others with kindness and respect.
You:

Project Wrap-Up

I did this project today to learn how to set up Amazon Bedrock to run Foundation Models and design multi-turn chatbot inside Amazon Cloudshell. Another skill I want to learn is how to create my own AI Foundation Model.

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